

Fiber intake and the risk of oral, pharyngeal and esophageal cancer.



The relation between various types of fiber and oral, pharyngeal and esophageal cancer was investigated using data from a case-control study conducted between 1992 and 1997 in Italy. Cases were 271 hospital patients with incident, histologically confirmed oral cancer, 327 with pharyngeal cancer and 304 with esophageal cancer. Controls were 1,950 subjects admitted to the same network of hospitals as the cases for acute, nonneoplastic diseases. Cases and controls were interviewed during their hospital stay using a validated food frequency questionnaire. Odds ratios (OR) were computed after allowance for age, sex, and other potential confounding factors, including alcohol, tobacco consumption, and energy intake. The ORs for the highest vs. the lowest quintile of intake of oral, pharyngeal and esophageal cancer combined were 0.40 for total (Englyst) fiber, 0.37 for soluble fiber, 0.52 for cellulose, 0.48 for insoluble non cellulose polysaccharide, 0.33 for total insoluble fiber and 0.38 for lignin. The inverse relation were similar for vegetable fiber (OR = 0.51), fruit fiber (OR = 0.60) and grain fiber (OR = 0.56), and were somewhat stronger for oral and pharyngeal cancer than for esophageal cancer. The ORs were similar for the two sexes and strata of age, education, alcohol and tobacco consumption, and total non-alcohol energy intake. Our study indicates that fiber intake may have a protective role on oral, pharyngeal and esophageal cancer.